

System Architecture & Production Design

GraphOne / FrontierAtlas — Global AI Intelligence Graph Platform

1. Scale Strategy: Ingesting 500,000+ Multi-Dimensional Records

Scaling ingestion to 500,000+ records without manual intervention requires transitioning from single-node asynchronous scrapers to a partitioned, event-driven distributed pipeline built on **Apache Kafka**, **Ray Core** worker pools, and an adaptive proxy/TLS impersonation mesh.

Pipeline Stage	Underlying Technology	Scaling & Concurrency Strategy
Frontier Scheduler	Apache Kafka + Redis	URL queues partitioned by domain hash (hash(host) % N). Domain-level rate limit isolation.
Async Crawler Mesh	Python asyncio + curl_cffi	JA3/JA4 TLS fingerprint spoofing; dynamic residential proxy mesh; automatic 403/429 IP eviction.
Heavy SPA Engine	Playwright Headless Pool	Browser context pooling for JS-rendered apps; intercepts raw GraphQL/XHR JSON responses directly.
LLM Extraction Pool	Ray Tasks + Central Redis Governor	Distributed token bucket throttler; semantic densifier preventing 413s; multi-tier fallback cascade.
Deterministic Entity Engine	Ray Cluster + Jaro-Winkler	Inverted index alias matching, legal suffix normalization, fuzzy token set resolution in <2ms/record.
Polyglot Storage	PostgreSQL + ClickHouse + Neo4j	Dual write: ACID relational metadata in Postgres, columnar time-series in ClickHouse, KG in Neo4j.

Core Pillars of 500k+ Data Acquisition

- **Domain-Partitioned Concurrency:** Isolating requests by target hostname prevents slow or throttled targets (e.g. ArXiv) from causing head-of-line blocking for fast targets (e.g. YC, Job boards).
- **TLS Impersonation & Anti-Bot Navigation:** Datadome and Cloudflare Turnstile actively detect Python aiohttp TLS signatures. Using curl_cffi impersonates genuine Chrome 128/Firefox TLS handshakes and HTTP/2 settings.
- **Automatic Dead-Letter Queue (DLQ):** Crawling failures that exceed max retries are automatically shunted to a Kafka DLQ for offline analysis and specialized headless replay.

2. Resilient LLM Extraction: Managing 413s & 429s at Scale

At thousands of concurrent extractions, LLM pipelines face strict rate envelopes (TPM/RPM) and payload context boundaries. Our multi-tier architecture prevents failures before they reach the provider.

A. Prevention of HTTP 413 (Payload Too Large)

- **DOM Stripping & Semantic Densification:** Pre-processing strips scripts, CSS, navigation bars, ads, and footers. Only semantic elements (<article>, <h1-h3>, <p>, meta tags) are preserved, reducing payload size by 75-90%.
- **Token Budget Governor:** Payload length is constrained to min(Model Context, 4000 tokens) - Prompt Reserve. Inputs exceeding this ceiling undergo sentence-boundary lookback truncation, preserving semantic integrity without mid-word cuts.
- **Hierarchical Map-Reduce:** Long multi-page documents (whitepapers, lengthy filings) are chunked into 2,000-token sections, processed in parallel, and consolidated through a structured merge step.

B. Resilience Against HTTP 429 (Rate Limits & Throttling)

- **Centralized Token Bucket Governor (Redis):** Workers reserve tokens prior to invocation against per-provider RPM/TPM sliding windows, stopping rate limit violations proactively.
- **Decorrelated Full Jitter Backoff:** When encountering a 429: delay = Uniform(0.5, 1.5) * min(MaxDelay, Base * 2^attempt). This spreads retry bursts uniformly, preventing synchronized thundering-herd spikes across workers.
- **Multi-Tier Failover Cascade:**

Tier	Provider / Model	Role & Failover Condition	Latency / Cost
Tier 1	Gemini 1.5 / 2.0 Flash	Primary high-throughput JSON extraction. Active provider.	<400ms / \$0.075/1M tok
Tier 2	Groq Llama 3.3 70B	Failover on Gemini 429, quota exhaustion, or timeout >4s.	<280ms / \$0.59/1M tok
Tier 3	DeepSeek Chat	Failover if Tier 1 & Tier 2 are saturated. Deep structural reasoning.	<750ms / \$0.14/1M tok
Tier 4	Deterministic Heuristic Parser	Zero-downtime safety net. Extracts schema via regex and DOM rules.	<2ms / \$0.00

3. Freshness Tracking: Zero-Duplicate Distributed Crawling

Guaranteeing that no article or job posting is processed twice within the 24-hour freshness window requires a multi-tier deduplication filter operating across memory, network, and database layers.

- **Canonical URL Normalization:** Strips query trackers (UTMs, fbclid, session IDs), removes trailing slashes, normalizes protocols, and resolves canonical redirects.
- **Distributed Bloom Filter (RedisBloom):** Sub-millisecond lookup checks seen URLs with <0.01% false positive probability across thousands of crawler nodes.
- **SimHash / MinHash Content Fingerprinting:** Syndicated articles often share identical text across different domains. 64-bit SimHash identifies semantic duplicates where Hamming Distance ≤ 3.
- **Idempotency Database Constraints:** PostgreSQL partial unique indices enforce: UNIQUE(canonical_entity_id, hash(job_title), publication_date::date).

4. Storage Strategy: Polyglot Multi-Model Intelligence Architecture

The GraphOne venture ecosystem exhibits four conflicting access patterns: transactional entity records, high-velocity signal streams, dense embedding vectors, and multi-hop relationship graphs. We implement a polyglot persistence architecture optimized for each workload.

Database Engine	Workload & Entities	Architectural Justification
PostgreSQL 16 (Primary Relational)	• Startups Core Records • Canonical Products • Entity Resolution Mappings • Pipeline Execution State	ACID compliance guarantees data integrity during concurrent entity resolution. JSONB columns accommodate rapidly evolving startup attributes without disruptive schema migrations. Connection pooling via PgBouncer.
ClickHouse (Real-Time Analytics)	• 24h News Signal Streams • Job Market Postings • GitHub Star Time-Series • Crawler Telemetry Logs	Columnar compression achieves up to 80% storage savings on repetitive strings. Ingests 250,000+ rows/sec per node. Sub-second analytical queries (e.g. <i>'Compute star velocity across 10,000 ML papers over 90 days'</i>).
Neo4j Enterprise (Graph Database)	• Intelligence Knowledge Graph • Founder-Startup-Product Links • Citation & Dependency Networks	Index-free adjacency provides constant-time $O(1)$ traversal per hop. Relational JOINS across 5 levels degrade exponentially; Neo4j executes complex graph pattern matching in milliseconds.
Qdrant / Pgvector (Vector Search)	• Entity Semantic Embeddings • Article & Paper Summaries • Fuzzy Semantic Matching	HNSW index enables semantic search: <i>'Find startups building autonomous synthetic data pipelines'</i> even without exact keyword overlaps.

5. Target Canonical Data Schemas

All ingested and resolved records adhere to strict, versioned schemas matching GraphOne specifications:

Startup Entity	schemaVersion, recordType='STARTUP', source.name, source.url, content.entityName, content.data.employeeCount, collectedAt
Product Entity	schemaVersion, recordType='PRODUCT', source.name, source.url, content.startupName, content.pricingModel (FREE FREEMIUM PAID ENTERPRISE), collectedAt
Research Paper	schemaVersion, recordType='RESEARCH_PAPER', content.title, content.authors, content.paper_url, content.github_url, content.github_stars, content.published_date
Job Entity	schemaVersion, recordType='JOB', source.name, content.company, content.date (<24h), content.is_remote, content.role_family

6. Compliance & Zero-Hallucination Guarantee

Every single record in the Intelligence Graph is grounded in an authentic source URL verified during ingestion. Raw HTTP responses and timestamps are preserved in immutable audit storage, guaranteeing complete lineage and reproducible trace-back for all 500,000+ entities.